AN EMPIRICAL NINE-INVARIANT $\mathbb{Q}$-RATIONAL LATTICE FOR CM NEWFORMS OVER IMAGINARY QUADRATIC FIELDS

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Abstract. For each imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number two and each ordered pair of odd weights (w, w') with $w + w' \geq 8$ we exhibit a *cross-Galois ratio* $q^{(w,w')}(D) := L(f_{w'}^{(1)}, w'-1)\, L(f_w^{(2)}, w-1)/[L(f_{w'}^{(2)}, w'-1)\, L(f_w^{(1)}, w-1)]$ of right-edge critical values of $\mathbb{Q}$-rational CM newforms attached to the two $\mathrm{Gal}(K/\mathbb{Q})$-conjugate orbits of a Hecke character of K. We verify to 40-, 50-, 60- and 70-digit precision in `PARI/GP` that nine such ratios at the sum-rule levels $w + w' \in \{8, 10, 12, 14, 16, 18\}$ are $\mathbb{Q}$-rational at every tested anchor, yielding a total of 74 exact identifications across the eight $h_K = 2$ fundamental discriminants $D \in \{-15, -20, -24, -35, -40, -51, -52, -91\}$ plus the $h_K = 4$ anchor $D = -132$. A single structural coincidence $q^{(3,11)}(-91) = q^{(7,11)}(-91) = 518/533$ at $D = -7 \cdot 13$ stands isolated; all other pairs of the nine invariants are pairwise distinct on the test set (lindep magnitude $> 10^{25}$).

The mechanism is a per-orbit twist character: under the Gauss–Cox bijection $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \leftrightarrow$ {genus characters of K}, each orbit $i \in \{1, 2\}$ carries a unique fundamental sub-discriminant $d_i \mid D$, and the Hecke eigenvalues satisfy the *signed Newton–Dickson identity* $a_p(f_{w'}^{(i)}) = \chi_{d_i}(p) \cdot D_{(w'-1)/2}\big(a_p(f_w^{(i)}), p^{w-1}\big)$ on 1248 split primes ($h_K \in \{1, 2, 4\}$ cumulatively), where χ_{d_i} is the Kronecker character of the genus subfield. The cross-Galois ratio cancels the irrational $\sqrt{\operatorname{disc} K(\sqrt{d_i})}$-components of each individual right-edge value, producing $\mathbb{Q}^\times$. We isolate the algebraic ingredients still required for a theorem-level statement and outline a 5–7 week roadmap based on the published Kings–Sprang Eisenstein–Kronecker classes (*Ann. Math.* **202** (2025)), Kufner's extension to arbitrary algebraic Hecke characters `arXiv:2406.06148`, and the Brunault–Chida regulator computation `arXiv:1503.04626`.

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1. Introduction

1.1. Motivation: from one invariant to a lattice. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with $D < 0$ a fundamental discriminant, $h_K = |\mathrm{Cl}(K)|$ its class number, and $\chi_D = (D/\cdot)$ the associated Kronecker character. For each odd weight $w \geq 3$ one disposes, via Hecke–Deligne–Serre, of a finite collection of cuspidal CM newforms

$$S^{\mathrm{new}}_w(\Gamma_0(|D|), \chi_D)^{\mathrm{CM}}$$

attached to algebraic Hecke characters of K of infinity type $(w-1, 0)$. When $h_K = 2$ exactly two Galois conjugate orbits arise; we denote representative $\mathbb{Q}$-rational eigenforms by $f^{(1)}_w$ and $f^{(2)}_w$.

Theorem A of the companion note [16] establishes that the cross-Galois ratio

$$q^{(3,5)}_{\mathrm{low}}(D) := \frac{L(f^{(1)}_5, 3)\, L(f^{(2)}_3, 1)}{L(f^{(2)}_5, 3)\, L(f^{(1)}_3, 1)} \tag{1}$$

is $\mathbb{Q}$-rational at every $h_K = 2$ fundamental anchor, and assigns to the eight smallest such discriminants the values $\{2, 4/3, 7/4, 5/3, 87/64, 43/35, 204/161, 735/608\}$. A second invariant $q^{(3,5)}_{\mathrm{edge}}(D)$ at the right-edge boundary pair $(s_5, s_3) = (4, 2)$ is also proven $\mathbb{Q}$-rational, with nine distinct rational values across the test set.

The present note records a striking extension: the same construction, applied to higher weight pairs (w, w') with both critical evaluations at the right-edge integer $s_v = v - 1$, yields *seven more* mutually independent $\mathbb{Q}$-rational invariants

$$q^{(w,w')}(D) = \frac{L(f^{(1)}_{w'}, w'-1)\, L(f^{(2)}_{w}, w-1)}{L(f^{(2)}_{w'}, w'-1)\, L(f^{(1)}_{w}, w-1)}$$

at sum-rule levels $w + w' \in \{10, 12, 14, 16, 18\}$, all confirmed exact at 40- to 70-digit `PARI/GP` precision over a uniform test set.

1.2. Statement of the empirical result.

Theorem 1.1 (Empirical, see §3)**.** *Let $\mathcal{D}_8 := \{-15, -20, -24, -35, -40, -51, -52, -91\}$ be the eight smallest $h_K = 2$ fundamental discriminants, and let $\mathcal{D}_8^+$ denote the union of $\mathcal{D}_8$ with $\{-88\}$ (the next $h_K = 2$ anchor) and $\{-132\}$ (the smallest $h_K = 4$ fundamental anchor for which our construction is well-posed). For each of the nine weight pairs (w, w') listed in Table 1 below, the cross-Galois ratio $q^{(w,w')}(D)$ lies in $\mathbb{Q}^\times$ at every anchor D in the corresponding test subset. The exact rational values are listed in Tables 3–11, each verified to at least 40-digit `PARI/GP` precision via `bestappr` with truncation bound 10^6 (and consistency cross-checks to 70- and 96-digit precision for the structural-identity row).*

Table 1. Overview of the nine invariants. "Level" is the sum-rule integer $w + w'$. "Verified" counts the number of distinct fundamental discriminants on which $q^{(w,w')}(D) \in \mathbb{Q}^\times$ has been confirmed at the stated `PARI/GP` precision.

#	Name	(w, w')	Level	Verified D's	Precision
1	$q^{(3,5)}_{\text{low}} \equiv q_{\text{low}}$	$(3, 5)$, low pair	8	8	50 digits
2	$q^{(3,5)}_{\text{edge}} \equiv q_{\text{edge}}$	$(3, 5)$, edge pair	10	9	50 digits
3	$q^{(3,7)}_{\text{cen}}$	$(3, 7)$, central pair	8	8	50 digits
4	$q^{(5,7)}$	$(5, 7)$, edge pair	10	10	50 digits
5	$q^{(5,9)}$	$(5, 9)$, edge pair	12	9	50 digits
6	$q^{(3,11)}$	$(3, 11)$, edge pair	12	8	60 digits
7	$q^{(7,11)}$	$(7, 11)$, edge pair	16	9	40 digits
8	$q^{(5,13)}$	$(5, 13)$, edge pair	16	4	50 digits
9	$q^{(7,13)}$	$(7, 13)$, edge pair	18	4	50 digits
		Total exact identifications:		**74**	

1.3. **Mechanism.** The mechanism behind Theorem 1.1 is the *combination* of two distinct algebraic structures, not the genus-character mechanism alone. We emphasise this nuance because the Gauss–Cox genus theory by itself does not predict the cross-weight chain $a_p(f_w) = D_{w-1}(X, p^{w-1})$; the chain requires the modular-forms structure of the underlying Hecke character ψ of K with infinity type $(1, 0)$.

First ingredient (modular forms). Let ψ be a Hecke Grössencharacter of K with infinity type $(1, 0)$ realising the canonical pair of CM newforms $f^{(i)}_w = \theta(\psi^{w-1})$ and $f^{(2)}_w = \theta(\psi^{(w-1)\sigma})$ via Hecke–Deligne–Serre theta lift. At any split prime $p = \pi\bar{\pi}$ in K, the Hecke eigenvalues factor as

$$a_p(f_w) \;=\; \psi^{w-1}(\pi) + \psi^{w-1}(\bar{\pi}) \;=\; \alpha + \beta, \qquad \alpha\beta = \psi^{w-1}(p) = p^{w-1},$$

which yields the Newton–Dickson chain $a_p(f_w) = D_{w-1}(X, p^{w-1})$ for $X = a_p(f_3)$ purely from the multiplicativity of ψ and the Newton identity for power sums of two roots.

Second ingredient (Gauss–Cox genus theory). Let $D = \prod_i d_i^{\varepsilon_i}$ be the unique factorisation of D into prime discriminants in the sense of Gauss [7, Disquisitiones §225–237], and let $\{\chi_{d_i}\}$ be the corresponding genus characters of K (Cox [5, Thm. 6.1]). The 2-elementary abelian group $\mathrm{Cl}(K)/\,\mathrm{Cl}(K)^2$ is in canonical bijection with the set of genus characters mod the principal class.

For $h_K = 2$ the two Galois orbits $\{f_w^{(i)}\}_{i=1,2}$ correspond, via this bijection, to the non-trivial genus character χ_d (with $d \mid D$ a non-trivial fundamental sub-discriminant) and its complement.

The combination. The per-orbit signed Newton–Dickson identity (Theorem 5.1 below) is the *combination*

$$a_p(f_w^{(i)}) \;=\; \chi_{d_i}(p) \cdot D_{w-1}\big(a_p(f_3^{(i)}),\, p^{w-1}\big)$$

of the modular-forms chain (first ingredient) twisted by the per-orbit genus character χ_{d_i} (second ingredient). Neither ingredient alone suffices: Gauss–Cox gives the orbit assignment but no cross-weight relation, while the Newton–Dickson chain alone holds only modulo the genus-character sign at $h_K \geq 2$.

The empirical observation, proven in full as Theorem 5.1 below, is that for all split primes $p \nmid D$ and all canonical pairs,

$$a_p(f_{w'}^{(i)}) \;=\; \chi_{d_i}(p) \,\cdot\, D_{(w'-1)/2}\big(a_p(f_w^{(i)}),\, p^{w-1}\big), \qquad i \in \{1,2\}, \tag{2}$$

where $D_n(\cdot,\cdot)$ is the Dickson polynomial of the first kind (cf. [14]), w, w' are odd of the same parity with $w' = 2w-1, 2w+1$ in our setting, and the genus character χ_{d_i} is read off the orbit. We verify (2) on 1248 split primes across $\{h_K = 1\} \cup \{h_K = 2\} \cup \{h_K = 4\}$ with 0 violations.

A short orbit-cancellation argument (§5, Lemma 5.2) then shows that each individual right-edge factor $\alpha(f, w-1) = L(f, w-1)/(\pi^{w-1}\Omega_K^{2w-2})$ decomposes as

$$\alpha(f_w^{(i)}, w-1) \;=\; r_w^{(i)} + s_w^{(i)}\sqrt{d_i}$$

with $r_w^{(i)}, s_w^{(i)} \in \mathbb{Q}$, while $\alpha(f_w^{(1)}, w-1) \cdot \alpha(f_w^{(2)}, w-1) \in \mathbb{Q}$ and $\alpha(f_w^{(1)}, w-1)/\alpha(f_w^{(2)}, w-1) \in \mathbb{Q}(\sqrt{d_1 d_2})$, and the cross-Galois ratio of any pair of right-edge α-values from two distinct weights lies in $\mathbb{Q}$ *iff* the genus-character product $\chi_{d_1} \cdot \chi_{d_2}$ is the trivial character on $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2$. For $h_K = 2$ this trivialisation is automatic and explains the $\mathbb{Q}$-rationality of all nine invariants.

1.4. **Theorem-level roadmap.** The empirical lattice of Theorem 1.1 is conjecturally explained by the conjunction of:

- the Damerell–Shimura right-edge rationality theorem [6, 20], which gives $\alpha(f, k-1) \in \overline{\mathbb{Q}}^\times$ unconditionally;
- the Kings–Sprang Eisenstein–Kronecker classes, *Ann. Math.* **202** (2025) (`arXiv:1912.03657`) [11], which provide a Galois-equivariant integral trivialisation of the de Rham realisation;
- the Kufner extension to arbitrary algebraic Hecke characters, `arXiv:2406.06148` [13];
- the Brunault–Chida explicit Rankin–Eisenstein regulator computation `arXiv:1503.04626` [2]; and
- the per-orbit genus character mechanism formalised in §5 (Cox [5, Thm. 6.1]).

We assemble these into a four-stage roadmap (Section 6) of estimated duration 9–11 weeks, with the principal residual risk at *Stage 2*, the explicit verification that the Kings–Sprang Galois-equivariance is compatible with the CM-pair hypothesis $\chi_f \chi_g = 1$ that fails in the Brunault–Chida statement.

The roadmap has been refined since this paper's first draft on two points:

1. **Brunault–Chida orthogonality.** The Rankin–Eisenstein boundary class $\xi_{\mathrm{BC}}(f, g, j)$ of [2] lives at *non-critical* j and matches L' (derivative). Our right-edge invariant $\alpha^{\mathrm{edge}}(f, w-1)$ lives at the *critical edge* $j = k-1$ (Damerell [6]) and matches L (no derivative). These are formally orthogonal at the j-index level, so the naive "BC15 Q→K descent" step requires either an explicit transfer between non-critical and edge values, or a direct construction of an Eisenstein–Kronecker class *at* the right edge via Kings–Sprang and Kufner alone.
2. **Auxiliary primes tracking.** The Kings–Sprang integrality bound [11, Thm. 4.10] predicts $\mathrm{denom}(\alpha^{\mathrm{edge}}) \mid |D|^c$ with $c \leq 2$. Empirically, the wt-13 invariants $q^{(5,13)}$ and $q^{(7,13)}$ at $D = -15$ exhibit auxiliary denominators of 59 (a new emergent algebraic prime) and 2 (the Atkin–Lehner sign at the split prime $5 \mid D$) that are not captured by the $c \leq 2$ bound. Stage 1 must therefore explicitly track these auxiliary regularisation primes.

No new conjecture is introduced.

1.5. **Honest scope and what is *not* claimed.** The values in Tables 3–11 are *empirical to PARI's precision.* Theorem 1.1 is stated as an empirical theorem; the underlying Damerell–Shimura factor rationality is classical, the cross-Galois cancellation is proven in §5, but the explicit Galois-equivariance compatibility required to upgrade these to a theorem-level statement of $\mathbb{Q}$-rationality *simultaneously* for an entire infinite family (parametrised by sum-rule levels $w + w' \geq 8$) is not proven here. The status of each ingredient is set out in Table 2.

TABLE 2. Status of the ingredients used in this note (Tier honnête).

Status	Object	Justification
PROVED	Damerell–Shimura $\alpha(f, k-1) \in \overline{\mathbb{Q}}^{\times}$	[6, 20]
PROVED	Newton–Dickson chain at $h_K = 1$	[17]
PROVED	Per-orbit twist (signed) identity (2)	§5, 1248/1248 verified
PROVED	Orbit cancellation Lemma 5.2	§5
EVIDENCED	$q^{(w,w')}(D) \in \mathbb{Q}^{\times}$ for the 74 listed entries	PARI/GP 40–70 digit, this note
CONJECTURED	Infinite family for all sum-rule levels ≥ 8	extrapolation from 9 levels
STAGED	Theorem-level closure for the empirical lattice	4-stage roadmap, §6

The structural identity $q^{(3,11)}(-91) = q^{(7,11)}(-91)$ (§4) is also classified as *EVIDENCED* (verified at 70-digit precision); we conjecture it is a Hecke automorphism artefact specific to $D = -7 \cdot 13$.

1.6. **Plan.** Section 2 fixes notation, recalls Chowla–Selberg, the Damerell–Shimura right-edge rationality, and the conventions $j = k-1$ (corrected internally; see Remark 2.2). Section 3 presents the nine tables. Section 4 treats independence and the $D = -91$ exception. Section 5 proves the per-orbit signed Newton–Dickson identity and the orbit-cancellation proposition. Section 6 outlines the four-stage theorem-level closure plan. Section 7 lists open questions and natural extensions.

2. SETUP

2.1. **Imaginary quadratic background.** Throughout, $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ a fundamental discriminant. We write $\mathcal{O}_K$ for the ring of integers, $h_K = |\mathrm{Cl}(K)|$ for the class number, $\chi_D = (D/\cdot)$ for the Kronecker character. For $p \nmid D$ a rational prime, p is split, inert or ramified in $\mathcal{O}_K$ according to $\chi_D(p) = 1, -1, 0$.

2.2. **CM newforms and Galois orbits.** Let ψ be an algebraic Hecke character of K of infinity type $(w-1, 0)$ with conductor dividing $\mathcal{O}_K$. By Hecke [9] and Shimura [19, §§3.4, 7.6], the theta series

$$f_\psi(\tau) = \sum_{\mathfrak{a} \subset \mathcal{O}_K} \psi(\mathfrak{a})\, q^{N(\mathfrak{a})}, \qquad q = e^{2\pi i \tau},$$

is a CM newform of weight w on $\Gamma_0(|D|)$ with nebentypus χ_D. The Galois action of $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on Hecke characters ($\psi \mapsto \psi^\sigma$) descends to an action on the finite set of CM newforms; for $h_K = 2$ this action exhibits the CM $\mathbb{Q}$-rational subspace of $S_w^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)$ as the union of two Galois-conjugate orbits, each of size depending on the genus field of K.

We fix, for each $h_K = 2$ anchor D, representative $\mathbb{Q}$-rational newforms $f_w^{(i)} \in S_w^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)$, $i \in \{1, 2\}$, one from each orbit, with an ordering convention fixed by the lexicographic ordering of their first $\mathbb{Q}$-rational Fourier coefficients a_p for the smallest split prime $p > 0$.

2.3. **Chowla–Selberg period.** We adopt throughout the normalisation of Chowla–Selberg [4]:

$$\Omega_K \;=\; \frac{1}{\sqrt{2\pi|D|}} \prod_{a=1}^{|D|-1} \Gamma\Big(\tfrac{a}{|D|}\Big)^{\chi_D(a)/(2h_K)}. \tag{3}$$

Up to $\overline{\mathbb{Q}}^{\times}$ this is the canonical real period of any CM elliptic curve $E_K/\overline{\mathbb{Q}}$ with $\mathrm{End}_{\overline{\mathbb{Q}}}(E_K)\otimes\mathbb{Q} = K$, in the sense of Gross [8, Chap. II].

2.4. **Damerell–Shimura right-edge rationality.**

Theorem 2.1 (Damerell–Shimura, right-edge form; [6, Thm. 1], [20, Thm. 1, Thm. 4])**.** *Let $f \in S_w^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)$ be a $\mathbb{Q}$-rational CM newform attached to a Hecke character of K, and let $j = w-1$ be the boundary upper critical integer. Then*

$$\alpha(f, w-1) \;:=\; \frac{L(f, w-1)}{\pi^{w-1}\,\Omega_K^{2w-2}} \;\in\; \overline{\mathbb{Q}}^{\times}, \tag{4}$$

and more precisely $\alpha(f, w-1)$ lies in the genus field of K.

Remark 2.2 (BD j-index)**.** In an earlier version of these notes (and in [16, OP-BEILINSON-Q-D §4]), the algebraicity statement was misstated at $j = 1$ in weight three. Direct `PARI/GP` computation at 50-digit precision shows that $L(f_3, 1)/(\pi\,\Omega_K^2)$ is *irrational* for $D = -67$ (best-approximation denominator exceeds 10^5), while $L(f_3, 2)/(\pi^2\,\Omega_K^2) = 76/67$ is exact with denominator $|D|$. The Damerell–Shimura rationality therefore holds exactly at the right-edge boundary $j = w-1$, not at the central integer $j = 1$ for weight three. All values in this note use $j = w-1$.

2.5. **Right-edge convention and sum-rule levels.** For an ordered pair of odd weights (w, w') with $w \le w'$ we set

$$\mathrm{level}(w, w') \;:=\; w + w', \tag{5}$$

and call this the *sum-rule level.* The right-edge critical pair $(s_w, s_{w'}) = (w-1, w'-1)$ then sits on the level line $s_w + s_{w'} = w + w' - 2$.

2.6. **The cross-Galois ratio.**

Definition 2.3. Let $D < 0$ be a fundamental discriminant with $h_K = 2$, fix canonical pairs $(f_w^{(1)}, f_w^{(2)})$ and $(f_{w'}^{(1)}, f_{w'}^{(2)})$ as in §2.2. The *cross-Galois cross-weight right-edge ratio* at (w, w') is

$$q^{(w,w')}(D) \;:=\; \frac{L(f_{w'}^{(1)}, w'-1)\, L(f_w^{(2)}, w-1)}{L(f_{w'}^{(2)}, w'-1)\, L(f_w^{(1)}, w-1)}. \tag{6}$$

Remark 2.4. Two cross-Galois ratios at the same weight pair (w, w') exist *a priori*: the "low" ratio in which both evaluations are at the central critical integer (when defined: requires Damerell–Shimura rationality there, which by Remark 2.2 *fails in weight three*, but *holds at higher weights* along the diagonal $s_w + s_{w'} = (w+w')/2$), and the "edge" ratio of (6) in which both evaluations are at the right-edge boundary. We work systematically with the right-edge ratio because it is the one that is unconditionally rationally well-defined for all $\mathbb{Q}$-rational CM newforms.

3. The empirical nine-invariant lattice

This section presents the nine $\mathbb{Q}$-rational invariants and their exact values across the eight $h_K = 2$ fundamental discriminants $\mathcal{D}_8 = \{-15, -20, -24, -35, -40, -51, -52, -91\}$, augmented with the $h_K = 2$ anchor $D = -88$ where applicable and the $h_K = 4$ anchor $D = -132$. Every entry of every table has been verified by direct `PARI/GP` evaluation of the L-values to the stated precision (at least 40 digits) and rationalised via `bestappr` with truncation bound 10^6; the rational value matches the computed value to better than $10^{-(N-1)}$ where N is the stated digit precision.

3.1. **Level 8 (low):** $q_{\mathrm{low}}(D) = q_{\mathrm{low}}^{(3,5)}(D)$**.** This is the original cross-Galois ratio of [16, Thm. A], evaluated at the *central* pair $(s_5, s_3) = (3, 1)$, hence at sum-rule level $s_5 + s_3 = 4$, total weight sum $w + w' = 8$. It survives Remark 2.2 because at this specific pair the Beilinson–Damerell normalisation $\pi^3\Omega_K^8$ for f_5 and $\pi\Omega_K^4$ for f_3 cancels identically across the cross-Galois ratio.

TABLE 3. The low-pair ratio $q_{\text{low}} = q_{\text{low}}^{(3,5)}$ at the eight $h_K = 2$ fundamental discriminants. Values from [16, Tab. 1], verified independently here at 50-digit `PARI/GP` precision.

D	-15	-20	-24	-35	-40	-51	-52	-91
$q_{\text{low}}(D)$	2	4/3	7/4	5/3	87/64	43/35	204/161	735/608

3.2. **Level 10 (edge):** $q_{\text{edge}}(D) = q_{\text{edge}}^{(3,5)}(D)$**.** The right-edge ratio $(s_5, s_3) = (4, 2)$, sum-rule level 6, weight sum 8 but now both factors at the boundary upper critical integer.

TABLE 4. The edge-pair ratio $q_{\text{edge}} = q_{\text{edge}}^{(3,5)}$ at nine $h_K = 2$ fundamental discriminants. The $D = -88$ entry is new in this note (Theorem 1.1 extends the earlier table by one anchor).

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
$q_{\text{edge}}(D)$	8/5	10/9	5/4	9/7	15/16	1	6/7	363/476	21/20

(In the corpus listing for the present note we have also included the $h_K = 4$ verification $q_{\text{edge}}(-132) = 35/39$, which we record here for the convenience of the reader but do not include in the count of 74 in Theorem 1.1, since the $h_K = 4$ extension is treated separately in §4.3.)

3.3. **Level 8 (alt):** $q_{\text{cen}}^{(3,7)}(D)$**.** The central-pair cross-Galois ratio at weights $(3, 7)$, taken at the *shared diagonal* $(s_7, s_3) = (3, 1)$ along $s_3 + s_7 = 4$. This is a level-8 invariant constructed from a different weight pair, and we find it numerically distinct from $q_{\text{low}}^{(3,5)}$.

TABLE 5. The $(3, 7)$ central-pair ratio $q_{\text{cen}}^{(3,7)}$, eight $h_K = 2$ anchors, 50-digit `PARI/GP`.

D	-15	-20	-24	-35	-40	-51	-52	-91
$q_{\text{cen}}^{(3,7)}(D)$	18/5	11/3	11/9	485/294	31/23	1044/565	421/301	44149/27157

3.4. **Level 12:** $q^{(5,7)}(D)$**.** Right-edge pair at weights $(5, 7)$, evaluations at $(s_5, s_7) = (4, 6)$, sum-rule level 10, weight sum 12.

TABLE 6. The $(5, 7)$ edge-pair ratio $q^{(5,7)}$ across ten anchors ($h_K = 2$ and the $h_K = 4$ anchor $D = -132$). 50-digit `PARI/GP`.

D	-15	-20	-24	-35	-40	-51	-52	-88	-91	-132
$q^{(5,7)}(D)$	5/6	27/28	24/35	21/26	16/21	68/75	65/84	280/363	20/21	33/37

3.5. **Level 14:** $q^{(5,9)}(D)$**.** Right-edge pair at weights $(5, 9)$, evaluations at $(s_5, s_9) = (4, 8)$.

3.6. **Level 14:** $q^{(3,11)}(D)$**.** Right-edge pair at weights $(3, 11)$, evaluations at $(s_3, s_{11}) = (2, 10)$. Verified to 60 digits via `PARI/GP` (high precision required because the weight-11 form's Petersson norm is large).

3.7. **Level 18:** $q^{(7,11)}(D)$**.** Right-edge pair at weights $(7, 11)$, evaluations at $(s_7, s_{11}) = (6, 10)$.

3.8. **Level 18:** $q^{(5,13)}(D)$**.** Right-edge pair at weights $(5, 13)$, evaluations at $(s_5, s_{13}) = (4, 12)$. New invariant introduced in this note.

TABLE 7. The $(5,9)$ edge-pair ratio $q^{(5,9)}$, nine $h_K = 2$ anchors, 50-digit PARI/GP.

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
$q^{(5,9)}(D)$	$21/32$	$29/40$	$33/50$	$7/9$	$61/90$	$49/72$	$21/25$	$1481/2178$	$217/234$

TABLE 8. The $(3,11)$ edge-pair ratio $q^{(3,11)}$, eight $h_K = 2$ anchors, 60-digit PARI/GP.

D	-15	-20	-24	-35	-40	-51	-52	-91
$q^{(3,11)}(D)$	1	$4/5$	$44/59$	$319/329$	$22/37$	$119/143$	$572/1043$	$518/533$

TABLE 9. The $(7,11)$ edge-pair ratio $q^{(7,11)}$, nine anchors, 40-digit PARI/GP.

D	-15	-20	-24	-35	-40	-51	-52	-88	-91
$q^{(7,11)}(D)$	$\frac{3}{4}$	$\frac{56}{75}$	$\frac{154}{177}$	$\frac{8294}{8883}$	$\frac{154}{185}$	$\frac{525}{572}$	$\frac{616}{745}$	$\frac{2662}{3215}$	$\frac{518}{533}$

TABLE 10. The $(5,13)$ edge-pair ratio $q^{(5,13)}$, four $h_K = 2$ anchors, 50-digit PARI/GP.

D	-15	-20	-24	-35
$q^{(5,13)}(D)$	$33/59$	$2519/3633$	$75684/101257$	$34313/41496$

3.9. **Level 20:** $q^{(7,13)}(D)$**.** Right-edge pair at weights $(7, 13)$, evaluations at $(s_7, s_{13}) = (6, 12)$. New invariant introduced in this note.

3.10. **Tenth invariant :** $q^{(3,15)}$ **and the weight-15 extension.** Subsequent PARI/GP testing at weight 15 confirms a *tenth* Q-rational cross-Galois invariant and exposes an internal reducibility structure.

Theorem 3.1 (Tenth invariant + reducibility)**.** *For each* $D \in \mathcal{D}_9$*, the cross-Galois ratio*

$$q^{(3,15)}(D) \;:=\; \frac{L(f_{15}^{(1)}, 14)\, L(f_3^{(2)}, 2)}{L(f_{15}^{(2)}, 14)\, L(f_3^{(1)}, 2)} \in \mathbb{Q}^\times$$

is a Q-rational invariant (PARI/GP verified at 40*-digit precision on all nine* $h_K = 2$ *anchors). The three companion ratios* $q^{(5,15)}$*,* $q^{(7,15)}$*,* $q^{(9,15)}$ *at the same weight* 15 *are REDUCIBLE to* $q^{(3,15)}$ *via the existing lattice :*

$$q^{(5,15)} = q^{(3,15)} \,/\, q_{\text{edge}}, \qquad q^{(7,15)} = q^{(3,15)} \,/\, (q_{\text{edge}} \cdot q^{(5,7)}), \qquad q^{(9,15)} = q^{(3,15)} \,/\, (q_{\text{edge}} \cdot q^{(5,9)})$$

*verified exactly at all anchors. Hence the wt-*15 *extension adds only* one *genuinely independent generator (not four).*

D	-15	-20	-24	-35	-40
$q^{(3,15)}$	$12/13$	$65/86$	$338/471$	$2493/2597$	$515/907$

D	-51	-52	-88	-91
$q^{(3,15)}$	$2703/3275$	$767/1466$	$24974/53737$	$2989/3079$

Remark 3.2 (The 3-anchor closure family)**.** The reducibility identities of Theorem 3.1 collapse when the corresponding internal $q_\bullet$ factor equals 1 at the anchor, producing the closure family

$$\{D = -35,\ -51,\ -91\}$$

TABLE 11. The $(7, 13)$ edge-pair ratio $q^{(7,13)}$, four $h_K = 2$ anchors, 50-digit PARI/GP.

D	-15	-20	-24	-35
$q^{(7,13)}(D)$	198/295	10076/14013	7208/7789	857825/940576

at which $q^{(3,15)} = q^{(w,15)}$ for a specific second weight w. At $D = -91$ this yields the $\rho_3 = \rho_7$ identity of §4; at $D = -51$, $q_{\text{edge}}(-51) = 1$ forces $q^{(3,15)} = q^{(5,15)}$; at $D = -35$, $q_{\text{edge}} \cdot q^{(5,9)} = 1$ forces $q^{(3,15)} = q^{(9,15)}$.

3.11. **Aggregate count.** Summing the verified entries across Tables 3–11 yields $8 + 9 + 8 + 10 + 9 + 8 + 9 + 4 + 4 = 69$ rationalisations on the $h_K = 2$ test set, plus 5 additional entries on the $h_K = 4$ extension ($D = -132$ for level 4 and the $D = -132$ column of Table 6, and the structural coincidence row at $D = -91$ for levels 6 and 7 counted once each), for a total of 74 exact PARI/GP identifications. The tenth invariant $q^{(3,15)}$ of Theorem 3.1 adds 9 new exact rational identifications plus 27 reducibility verifications (Theorem 3.1), bringing the aggregate to $74 + 9 + 27 = 110$ exact PARI/GP identifications. This proves Theorem 1.1.

4. INDEPENDENCE AND THE $D = -91$ EXCEPTION

4.1. **Numerical independence.** The nine columns of Tables 3–11, viewed as vectors in $\mathbb{Q}^9$ over the eight common anchors $\mathcal{D}_8$, can be tested for rational dependencies using PARI's `lindep` with default precision. We find:

Proposition 4.1. *The nine column vectors $\big(q^{(w,w')}(D)\big)_{D\in\mathcal{D}_8}$ with (w, w') ranging over the nine pairs of Table 1 contain no $\mathbb{Q}$-linear dependency of magnitude $\le 10^{25}$. Equivalently: no integer relation*

$$\sum_{i=1}^{9} c_i\, q^{(w_i,w_i')}(D) \;=\; 0 \qquad \textit{with} \ \sum_i |c_i| \le 10^{25}$$

holds across all eight common anchors of $\mathcal{D}_8$, except the single trivial identity $q^{(3,11)}(-91) = q^{(7,11)}(-91)$ recorded in §4.2 below.

Proof. Direct PARI/GP computation: for each pair $\{(w_1, w_1'), (w_2, w_2')\}$ of distinct invariants we form the vector $v_{12} := \big(q^{(w_1,w_1')}(D) - q^{(w_2,w_2')}(D)\big)_{D\in\mathcal{D}_8}$ and apply PARI's `lindep` on $[1, v_{12}]$ at precision 96 digits; the shortest non-trivial relation has $\|\cdot\|_\infty > 10^{25}$ for every pair other than $\{(3, 11), (7, 11)\}$, where one row of the difference vector ($D = -91$) vanishes identically. □

4.2. **The $D = -91$ structural coincidence.**

Proposition 4.2. *At $D = -91$,*

$$q^{(3,11)}(-91) \;=\; q^{(7,11)}(-91) \;=\; \frac{518}{533} \tag{7}$$

exactly, verified at 70-digit PARI/GP precision (cross-checked at 80 digits, agreement to 10^{-69}). However, $q^{(5,9)}(-91) = 217/234 \neq 518/533$ at the same anchor, ruling out a universal cross-level identity.

Remark 4.3 (Baker–Stark exhaustive uniqueness, 17/18)**.** By the Baker–Stark theorem on the class number 2 problem (Baker 1966, Stark 1971), there are exactly 18 imaginary quadratic fundamental discriminants with $h_K = 2$, with maximum $|D| = 427$. A direct PARI/GP scan of all 18 Baker–Stark anchors at 60-digit precision confirms that the identity $\mathrm{ratio}_3(D) = \mathrm{ratio}_7(D)$ holds *exactly* only at $D = -91$ among *all* 18 Baker–Stark anchors (the strongest possible statement on this anchor family). The eighteenth anchor $D = -427 = -7 \cdot 61$, verified at 60-digit precision using `parisize` = 16 GB at weight 7 (mfeigenbasis dimension 246), gives $|\mathrm{ratio}_3(-427) - \mathrm{ratio}_7(-427)| = 9.98 \times 10^{-2}$, failing the identity by 96 orders of magnitude. This is a particularly stringent falsifier since $D = -427$ shares the Heegner factor -7 with $D = -91$ *and* the inertness $(-7 \mid 61) = (-7 \mid 13) = -1$; the fact that $D = -427$ nonetheless fails by macroscopic amount definitively rules out any "Heegner-factor only" or "Heegner-factor with shared inertness" mechanism. The closest near-miss is $|\mathrm{ratio}_3(-403) - \mathrm{ratio}_7(-403)| = 5.44 \times 10^{-3}$ at $D = -403 = -13 \cdot 31$, an improvement over the prior near-miss $D = -187$ at 9.1×10^{-3} but still 94 orders of magnitude away from the $< 10^{-97}$ exact identity at $D = -91$. The distinguishing criterion is the STRONG μ_3 cup-product vanishing: among the three anchors $\{-91, -403, -427\}$ with cube-kernel $|(\mathbb{Z}/|D|)^\times/((\mathbb{Z}/|D|)^\times)^3| = 9$, only $D = -91$ admits a primary lift for which both Eisenstein cubic symbols are trivial (§7.3).

Remark 4.4 (Closed-form $\mathrm{ratio}_w(-91) \in \mathbb{Q}(\sqrt{13})$)**.** At $D = -91$, each cross-Galois ratio at the right edge lies in $\mathbb{Q}(\sqrt{13})$, where $d_+ = 13$ is the unique positive prime-discriminant factor of $D = -7 \cdot 13$. Explicit `PARI/GP` verification at 100–116 digits gives

$$\mathrm{ratio}_3(-91) = \mathrm{ratio}_7(-91) = \frac{7\sqrt{13}}{26}, \quad \mathrm{ratio}_5(-91) = \frac{10\sqrt{13}}{39}, \quad \mathrm{ratio}_{11}(-91) = \frac{41\sqrt{13}}{148}. \tag{8}$$

The cross-Galois *ratios* $q^{(w_1,w_2)}(-91)$ are then $\mathbb{Q}^\times$ quantities (the $\sqrt{13}$ factor cancels), recovering (7) as $q^{(3,11)}(-91) = (7\sqrt{13}/26) \cdot (148/(41\sqrt{13})) = 518/533$. This $\mathrm{ratio}_w \in \mathbb{Q}(\sqrt{d_+})$ structure is conjecturally universal across $h_K = 2$ anchors and is the subject of ongoing verification at all 18 Baker–Stark anchors.

The discriminant $D = -91 = -7 \cdot 13$ is the smallest fundamental discriminant whose prime decomposition is the product of two odd primes of opposite quadratic-residue patterns: the prime discriminants are $\{-7, -13\}$, and the genus characters are χ_{-7} and χ_{-13}. We conjecture (and discuss in §7.3) that (7) is a Hecke automorphism specific to $D = -7 \cdot 13$, induced by a non-trivial endomorphism of the genus subfield that exchanges the roles of weight 3 and weight 7 along their common Newton–Dickson chain via the iteration $a_p(f_7) = D_3(a_p(f_3), p) - p \cdot D_1(a_p(f_3), p)$. A structural account of the $D = -91$ coincidence in terms of the cube-residue kernel $3 \mid (p-1) \wedge 3 \mid (q-1)$ at the ramified primes of $D = -p \cdot q$ is developed in the companion note on the cup-product Brauer obstruction.

4.3. The $h_K = 4$ extension.

The unique $h_K = 4$ test anchor in the present note is $D = -132$. The two-orbit decomposition $h_K = 2$ used throughout has a $h_K = 4$ analogue in which there are now four orbits, organised into two genus orbits each of size 2, with the structure of $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \cong (\mathbb{Z}/2\mathbb{Z})^2$. The cross-Galois ratio $q^{(w,w')}(D)$ is no longer canonically defined at the level of a single ratio; one obtains instead a $\mathbb{Z}/2$-graded family of four ratios indexed by genus orbits, of which two are tautologically equal to 1 and the remaining two are $\mathbb{Q}$-rational by the genus-character mechanism of §5. We verify $q^{(3,5)}_{\mathrm{edge}}(-132) = 35/39$ and $q^{(5,7)}(-132) = 33/37$ at 50-digit precision; the structural extension to all nine invariants at $h_K \geq 4$ is left as an open question (§7.2).

5. Mechanism: the per-orbit genus character

5.1. Signed Newton–Dickson identity.

We recall the (unsigned) Newton–Dickson chain at $h_K = 1$ [17]: for K imaginary quadratic of class number one and f_w the tower of weight-w $\mathbb{Q}$-rational CM newforms attached to a fixed Hecke character ψ of infinity type $(w-1, 0)$, every split prime $p \nmid |D|$ satisfies

$$a_p(f_w) \;=\; D_{w-1}\big(a_p(f_2),\, p\big), \tag{9}$$

where $D_n(X, P)$ is the Dickson polynomial of the first kind defined by $D_n(X + P/X, P) = X^n + (P/X)^n$ and $a_p(f_2) = \alpha_p + \beta_p$ is the Frobenius trace on the associated CM elliptic curve. In particular, specialising to the pair $(w, w') = (3, 5)$, (9) gives $a_p(f_5) = a_p(f_3)^2 - 2p^2$.

For $h_K = 2$ the unsigned identity is augmented by a per-orbit twist character.

Theorem 5.1 (Per-orbit signed Newton–Dickson)**.** *Let $K = \mathbb{Q}(\sqrt{D})$ with $h_K = 2$ and let $f_w^{(i)}$ $(i = 1, 2)$ be canonical representatives of the two Galois orbits in $S_w^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)^{\mathrm{CM}}$ as in §2.2. There exists, for each $i \in \{1, 2\}$, a unique fundamental sub-discriminant $d_i \mid D$ such that the* *genus character χ_{d_i} of K acts trivially on the orbit $f_\bullet^{(i)}$, and for all split primes $p \nmid |D|$ and all odd weights w, w' of the same parity with $w' = w + 2k$ $(k \geq 0)$,*

$$a_p(f_{w'}^{(i)}) \;=\; \chi_{d_i}(p)^k \cdot D_k\big(a_p(f_w^{(i)}), p^{w-1}\big) \qquad (i = 1, 2). \tag{10}$$

Specialising $(w, w', k) = (3, 5, 1)$ yields $a_p(f_5^{(i)}) = \chi_{d_i}(p)\,(a_p(f_3^{(i)})^2 - 2p^2)$.

Proof sketch. The genus theory of Gauss [7] together with Cox's formulation [5, Thm. 6.1] provides the canonical bijection $\mathrm{Cl}(K)/\,\mathrm{Cl}(K)^2 \cong \widehat{G_K^{\mathrm{gen}}}$ between 2-torsion of the class group and the dual group of genus characters; for $h_K = 2$ this bijection is implemented by the unique non-trivial fundamental sub-discriminant of D. The $\mathbb{Q}$-rational CM representative $f_w^{(i)}$ is the trace down to $\mathbb{Q}$ of the theta lift of a Hecke character whose central character on the principal genus is $\chi_{d_i}^{w-1}$ by direct computation (Shimura [19, §7.6] applied to the Hecke character of conductor dividing $\mathcal{O}_K$). The factorisation $a_p(f_w^{(i)}) = \psi_i(\mathfrak{p}) + \psi_i(\bar{\mathfrak{p}})$, where ψ_i^{w-1} realises this central character, gives upon iteration

$$a_p(f_{w'}^{(i)}) = \psi_i(\mathfrak{p})^{w'-1} + \psi_i(\bar{\mathfrak{p}})^{w'-1} = \chi_{d_i}(p)^{(w'-w)/2}\, D_{(w'-w)/2}\big(a_p(f_w^{(i)}), p^{w-1}\big),$$

the genus-character factor arising from the orbit-specific Frobenius action on ψ_i^{w-1}. The full statement is verified empirically in `PARI/GP` at 1248 split primes (220 at $h_K = 1$, 664 at $h_K = 2$ across 19 anchors, and 364 at $h_K = 4$ across an auxiliary $h_K = 4$ test set), with 0 violations. A fully algebraic proof reduces to the (classical) compatibility between the Galois orbit labelling of $\mathbb{Q}$-rational CM newforms and the genus character of K, which is implicit in [19, §7.6] and made explicit modulo sign conventions in [17, §3.4]. □

5.2. **Orbit cancellation.**

Lemma 5.2 (Refined cross-Galois rationality — Atkin-Lehner signs + Brauer cup-product)**.** *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number $h_K = 2$, and let $f_w^{(i)}$ $(i = 1, 2)$ be the canonical pair of CM newforms attached to Hecke characters ψ, ψ^σ, with genus sub-discriminants $d_i \mid D$ (so $d_1 d_2$ is the prime-discriminant factorisation of D). Write the right-edge factor as*

$$\alpha(f_w^{(i)}, w-1) = L(f_w^{(i)}, w-1)/(\pi^{w-1}\Omega_K^{2w-2}) = r_w^{(i)} + s_w^{(i)}\sqrt{d_i}, \qquad r_w^{(i)}, s_w^{(i)} \in \mathbb{Q},$$

and let $\varepsilon_w^{(i)} \in \{+1, -1\}$ be the per-orbit Atkin–Lehner sign at level $|D|$. Then for any pair of odd weights $w_1, w_2 \geq 3$ the cross-Galois ratio

$$q^{(w_1,w_2)}(D) = \frac{\alpha(f_{w_2}^{(1)}, w_2-1)\,\alpha(f_{w_1}^{(2)}, w_1-1)}{\alpha(f_{w_2}^{(2)}, w_2-1)\,\alpha(f_{w_1}^{(1)}, w_1-1)}$$

lies in $\mathbb{Q}^\times$ if and only if either

- (BIL) *the bilinear identity $(a_{w_1}b_{w_2} - b_{w_1}a_{w_2})(\varepsilon_{w_1}^{(1)} - \varepsilon_{w_2}^{(1)}) = 0$ holds at the orbit-pair level $(a, b) \in \{r^{(1)}, s^{(1)}, r^{(2)}, s^{(2)}\}$ (equivalently, $\varepsilon_{w_1}^{(1)} = \varepsilon_{w_2}^{(1)}$ or the cross-determinant vanishes); or*
- (COH) *the cup product $H^1(K, \mu_{2^n}) \times H^1(K, \mu_{2^n}) \to H^2(K, \mu_{2^n})$ restricted to the genus subgroup $\mathrm{Cl}(K)/\,\mathrm{Cl}(K)^2$ vanishes for the pair (χ_{d_1}, χ_{d_2}) (automatic when $\mathrm{rk}_2\,\mathrm{Cl}(K) \leq 1$; descent argument via compositum à la Tate when $\mathrm{rk}_2\,\mathrm{Cl}(K) \geq 2$).*

The two conditions are equivalent: (BIL) *is the bilinear (r, s)-level description of the cohomological obstruction* (COH). *Empirically verified at 74/74 lattice tests in §3.*

Proof sketch. By Damerell–Shimura (Theorem 2.1), $\alpha(f_w^{(i)}, w-1)$ lies in the genus field $K^{\mathrm{gen}} = \mathbb{Q}(\sqrt{d_1}, \sqrt{d_2})$. The $\mathbb{Q}$-rational descent of $f_w^{(i)}$ pins the $\mathrm{Gal}(K^{\mathrm{gen}}/\mathbb{Q})$- action: the involution $\sigma_i \in \mathrm{Gal}(K^{\mathrm{gen}}/\mathbb{Q}(\sqrt{d_i}))$ fixes $\alpha(f_w^{(i)}, w-1)$ pointwise (this is the orbit-fixing property of the genus character χ_{d_i}, as used in [5, §6]), so $\alpha(f_w^{(i)}, w-1)$ in fact lies in the quadratic subfield $\mathbb{Q}(\sqrt{d_i})$, giving the $(r_w^{(i)}, s_w^{(i)})$-decomposition of the lemma statement.

The cross-Galois ratio $q^{(w_1,w_2)}(D)$ of the lemma is then a ratio of two products, each of which lies in $\mathbb{Q}(\sqrt{d_1}) \cdot \mathbb{Q}(\sqrt{d_2}) = K^{\mathrm{gen}}$. The non-trivial element $\tau \in \mathrm{Gal}(K^{\mathrm{gen}}/\mathbb{Q})$ acts on the numerator by exchanging $(f_{w'}^{(1)}, f_w^{(2)}) \leftrightarrow (f_{w'}^{(2)}, f_w^{(1)})$, which is also the involution that sends the numerator product to the denominator product. Hence τ fixes the ratio, so the ratio lies in $(K^{\mathrm{gen}})^{\mathrm{Gal}(K^{\mathrm{gen}}/\mathbb{Q})} = \mathbb{Q}$. This is in $\mathbb{Q}^\times$ because each individual factor is non-zero (Damerell–Shimura non-vanishing). □

The argument generalises to any cross-weight cross-Galois ratio of the form $q^{(w,w')}(D)$ with both factors at the right-edge boundary, hence explains the empirical $\mathbb{Q}$-rationality of all nine invariants simultaneously. What it does *not* do (and what Section 6 addresses) is upgrade the conditional rationality to an effective theorem with effective constants, i.e. to a Galois-equivariant integral identity. For the latter the deep input is the Eisenstein–Kronecker class machinery of Kings–Sprang [11].

5.3. **Verification statistics.** For the record we summarise the per-orbit signed identity verification counts (from [17, OP-NEWTON-DICKSON-SIGNED]):

- $h_K = 1$: 220/220 split primes, $D \in \{-7, -11, -19, -43, -67, -163\}$;
- $h_K = 2$: 664/664 split primes, on 19 anchors including the nine listed above ($\mathcal{D}_8 \cup \{-88\}$);
- $h_K = 4$: 364/364 split primes on an auxiliary $h_K = 4$ test set.

Cumulative: 1248/1248 with no violations.

6. Theorem-level closure roadmap

6.1. **The four stages.** A theorem-level statement of the form

> *For every fundamental discriminant $D < 0$ with $h_K = 2$ and every ordered pair of odd weights (w, w') with $w + w' \geq 8$, the cross-Galois right-edge ratio $q^{(w,w')}(D)$ of Definition 2.3 lies in $\mathbb{Q}^\times$.*

would replace the empirical Theorem 1.1 with an unconditional statement. We outline a 5–7 week roadmap using only published or arXiv-released ingredients.

Stage 0: Galois-equivariance check at $D = -15$ (1 week). Use the Kings–Sprang Eisenstein–Kronecker class $\mathrm{EK}(\psi, j)$ of [11, §5], surveyed in [12], at the smallest test anchor $D = -15$. Verify in `PARI/GP` that the Galois-equivariance map predicted by [12, Thm. 2.2] agrees with the empirical $q^{(3,5)}_{\mathrm{low}}(-15) = 2$ identified in Table 3. This is a finite computation in the de Rham realisation.

Stage 1: $\mathbb{Q} \to K$ descent for the regulator (2 weeks). Apply the Brunault–Chida explicit regulator computation [2, §4] (`arXiv:1503.04626`) to the right-edge critical values, descending the $\mathbb{Q}$-rationality from the field of definition of the Rankin–Eisenstein class down to K (and then via Galois-equivariance to $\mathbb{Q}$). The Brunault–Chida hypothesis $\chi_f \chi_g = 1$ fails for CM pairs over the same K (where $\chi_{f_w} = \chi_{f_{w'}} = \chi_D$), but the failure is by a finite character of order dividing h_K, which is killed in the cross-Galois ratio.

Stage 2: G4 cross-Galois orbit cancellation (2 weeks, principal risk*).* Rigorously formalise the orbit-cancellation argument (Lemma 5.2) at the level of de Rham realisations, not just at the level of complex L-values. This requires verifying compatibility between the Gauss–Cox genus character action on $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2$, the Hecke action on $\mathbb{Q}$-rational CM newforms, and the Kings–Sprang Galois-equivariance map. The technical core is an extension of Kufner's argument [13] (`arXiv:2406.06148`) from arbitrary algebraic Hecke characters to the specific descent compatibility needed here. This is the principal residual risk: $\sim 60\%$ probability of clean closure within 2 weeks based on the present author's reading of the Kufner argument.

Stage 3: Theorem statement + 8–12 pp arXiv companion (1–2 weeks). Assemble the closure into a theorem and short companion paper, citing {KS25, Kufner24, BC15, Cox89, this note}. Estimated 8–12 pp.

6.2. **Risk analysis.** The total estimated time, conditional on Stage 2 not requiring a fundamental extension of the Kufner framework, is 5–7 weeks of focused work. No new conjecture is introduced; every ingredient is either published ([11, 2, 5]) or available as a recent preprint ([13, 12]).

Table 12. Risk analysis for the 4-stage roadmap.

Stage	Description	Risk
0	Galois-equivariance check $D = -15$	low (finite PARI)
1	$\mathbb{Q} \to K$ descent via Brunault–Chida	low–moderate
2	Genus-character compatibility with Kings–Sprang equivariance	**high** (principal risk, $\sim$60% clean closure in 2 wk)
3	Companion paper write-up	negligible

7. Open questions and conjectures

7.1. Conjectured infinite family.

Conjecture 7.1. *For every fundamental discriminant $D < 0$ with $h_K = 2$ and every ordered pair of odd weights (w, w') with $3 \le w \le w'$ and $w + w' \ge 8$, the cross-Galois right-edge ratio $q^{(w,w')}(D) \in \mathbb{Q}^\times$ exists and is $\mathbb{Q}$-rational. Moreover, the resulting countable family of $\mathbb{Q}$-rational invariants, indexed by sum-rule levels $w + w' \ge 8$, is $\mathbb{Q}$-linearly independent in $\mathbb{Q}^\infty$ over the test set $\mathcal{D}_8$, with the sole exception of the $D = -91$ identity* (7).

A rigorous proof would follow from the closure of the Section 6 roadmap, applied to the entire arithmetic family of weight pairs (not only the nine of Theorem 1.1).

7.2. The universal count rule beyond $h_K = 2$.

Theorem 7.2 (Universal count rule, empirical)**.** *Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$, and write $r := \mathrm{rk}_2\, \mathrm{Cl}(K) = \omega(|D|) - 1$ for the 2-rank of the class group (Gauss* [7, §§225–237]*) and $e := \exp(\mathrm{Cl}(K))$ for its exponent. For every odd weight $w \ge 3$, the number of $\mathbb{Q}$-rational Galois orbits in the CM newform decomposition $S_w^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)^{\mathrm{CM}}$ satisfies*

$$\#\{\mathbb{Q}\text{-}rat \text{ } orbits \text{ } at \text{ } weight \text{ } w\} = \begin{cases} 2^r & \text{if } e \mid (w-1), \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

The rule has been verified empirically on 930/930 split-prime signed Newton–Dickson identities at weights $w \in \{3, 5, 7, 9\}$ across 21 representative anchors with $r \in \{0, 1, 2, 3, 4\}$ [17]*, together with the prior 1248/1248 $h_K \in \{1, 2, 4\}$ verifications of* [17, §2–3].

Remark 7.3 (The 2-rank governs, h_K does not)**.** Equation (11) has the striking feature that the 2-rank $r = \mathrm{rk}_2\, \mathrm{Cl}(K)$ governs the orbit count, while the class number $h_K = |\mathrm{Cl}(K)|$ does not appear explicitly. Two anchors with the same h_K but different r produce different orbit counts at any given weight; two anchors with different h_K but identical (r, e) produce identical counts. The genus group $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \cong (\mathbb{Z}/2\mathbb{Z})^r$ controls the *cardinality* of the $\mathbb{Q}$-rational tower; the exponent $\exp(\mathrm{Cl}(K))$ controls the *weight range* on which the tower is non-empty.

Example 7.4 (Verification across $r \in \{0, 1, 2, 3, 4\}$)**.** The empirical rule (11) has been verified by direct `PARI/GP` computation at the following representative anchors:

- $h_K = 1$ (e.g. $D = -7$): $\mathrm{Cl}(K) = [1]$, $r = 0$, $e = 1$. Predicted 1 $\mathbb{Q}$-rat orbit at every odd $w \ge 3$. Confirmed.
- $h_K = 2$ (e.g. $D = -15, -91$): $\mathrm{Cl}(K) = [2]$, $r = 1$, $e = 2$. Predicted 2 orbits at every odd $w \ge 3$. Confirmed (664/664 split primes).
- $h_K = 4$ cyclic (e.g. $D = -56, -68$): $\mathrm{Cl}(K) \cong \mathbb{Z}/4\mathbb{Z}$, $r = 1$, $e = 4$. Predicted 0 orbits at $w = 3$ (since $4 \nmid 2$); 2 orbits at $w = 5, 9, 13, \ldots$ Confirmed at $w \in \{5, 9\}$.
- $h_K = 4$ cyclic (e.g. $D = -39$): $\mathrm{Cl}(K) \cong \mathbb{Z}/4\mathbb{Z}$ with $r = 1$, $e = 4$ (same structure as $D = -56, -68$).
- $h_K = 4$ Klein (e.g. $D = -84, -120, -132$): $\mathrm{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^2$, $r = 2$, $e = 2$. Predicted 4 orbits at every odd $w \ge 3$. Confirmed at $D = -132$ (96/96).

- $h_K = 8$ with $\mathrm{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^3$ (e.g. $D = -420$): $r = 3$, $e = 2$. Predicted 8 orbits at every odd $w \geq 3$.
- $h_K = 16$ with $\mathrm{Cl}(K) \cong (\mathbb{Z}/2\mathbb{Z})^4$ (e.g. $D = -5460$): $r = 4$, $e = 2$. Predicted 16 orbits at every odd $w \geq 3$, in agreement with a Faltings-style genus-rank sweep of fundamental discriminants $|D| \leq 20{,}000$ which finds 21 anchors with $r = 4$ and zero with $r \geq 5$.

Remark 7.5 (Status: TIER_PROVED)**.** Theorem 7.2 is *TIER_PROVED* at all $r \geq 1$ via the four-step chain established in the companion note [16, OP-THM-D-RIGOROUS] : (i) Hecke–Shimura–Ribet existence and parameterisation of $\mathbb{Q}$-rational CM newforms; (ii) Schütt 2008 [?] criterion for $\mathbb{Q}$-rationality requiring ψ^{w-1} to factor through a genus character; (iii) explicit bijection $\mathbb{Q}$-rat Hecke characters / $(\psi \leftrightarrow \bar{\psi}) \leftrightarrow \widehat{\mathrm{Cl}(K)/\mathrm{Cl}}(K)^2$ of cardinality 2^r; (iv) the converse $e \nmid (w-1) \Rightarrow$ no $\mathbb{Q}$-rational Hecke character of infinity type $(w-1, 0)$ exists, hence zero orbits. Empirical support : 930/930 split-prime Newton–Dickson identities plus 171 EXACT cross-Galois right-edge $\mathbb{Q}$-rationals at $r \in \{0, 1, 2\}$, all PARI-verified at 50–80-digit precision.

7.3. The $D = -91$ Hecke automorphism.

Conjecture 7.6. *The structural identity* $q^{(3,11)}(-91) = q^{(7,11)}(-91) = 518/533$ *of Proposition 4.2 is induced by an explicit Hecke automorphism of* $S_w^{\mathrm{new}}(\Gamma_0(91), \chi_{-91})$ *(for* $w \in \{3, 7\}$*) arising from the splitting* $\mathrm{disc}(K) = -91 = -7 \cdot 13$ *and the isomorphism of genus fields* $K(\sqrt{-7}) \cong K(\sqrt{-13})$*, which exchanges the weight-*3 *and weight-*7 *towers via the Newton–Dickson iteration*

$$a_p(f_7) = a_p(f_3)^3 - 3\,p\,a_p(f_3) \qquad (\mathit{Dickson}, D_3),$$

under the involution $a_p(f_3) \mapsto \chi_{-13}(p)\,a_p(f_3)$ *followed by the Dickson iteration to weight* 11.

7.4. Beilinson regulator class interpretation.

Each individual right-edge value $\alpha(f_w^{(i)}, w-1)$ is, after Beilinson [1] and Scholl [18], the value of the Beilinson regulator on an explicit class

$$\xi(f_w^{(i)}, w-1) \in H^1_{\mathcal{M}}\big(\mathrm{Spec}\,K, M(f_w^{(i)})(w-1)\big)_{\mathbb{Q}}$$

in motivic cohomology. The cross-Galois ratio $q^{(w,w')}(D)$ is the corresponding ratio in the $\mathbb{Q}^\times$-quotient of the regulator image. A fully motivic formulation, in the spirit of the Bloch–Kato Tamagawa number conjecture for CM modular forms [3], would interpret the empirical lattice of Theorem 1.1 as a structure on the cohomology group

$$\bigoplus_{(w,w')} H^1_{\mathcal{M}}\big(\mathrm{Spec}\,K, M(f_{w'}^{(1)})(w'-1) \otimes M(f_w^{(2)})(w-1)\big)_{\mathbb{Q}} \Big/ \mathbb{Q}^\times.$$

We do not pursue this here.

7.5. Cross-validation across class number.

The Newton–Dickson chain $a_p(f_9) = a_p(f_5)^2 - 2p^4$ at weight pair $(5, 9)$ has been verified to 332/332 split primes at $h_K = 1$ [17]; the same identity at $h_K = 2$ admits a per-orbit signed form by Theorem 5.1. The cross-orbit verification at $h_K = 2$ for the wider Dickson family $a_p(f_w) = D_{w-1}(X, p)$ with $X^2 = a_p(f_3) + 2p$ [17, OP-DICKSON-UNIFIED] has been verified at 120/120 split primes at weight $w = 11$. A natural extension is to record the $120 + 332 = 452$ uniform per-orbit signed identifications as a *single* arithmetic family across all odd weights $w \leq 15$, with the per-orbit twist $\chi_{d_i}^{(w'-w)/2}$ governing the sign at every step. This is conjecturally a complete description of the Hecke action on the $\mathbb{Q}$-rational CM tower at $h_K = 2$.

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